British Math Olympiad Round 1 2002 — P4/5

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Twelve people are seated around a circular table. In how many ways can six pairs of people engage in handshakes so that no arms cross? (Nobody is allowed to shake hands with more than one person at once.)

Solution

We will solve the following more general problem: Let n be a positive integer. Given $2n$ people are sat around a circular table, in how many ways can n pairs of people engage in handshakes so that no arms cross?

Let's denote the $2n$ people sat at the table by $a_1, a_2, ..., a_{2n}$, where each of the people are sat in that order.

Lemma: If person a_k handshakes with person a_{k+d} then each of the people $a_{k+1}, a_{k+2}, ..., a_{k+d-1}$ must handshake another person amongst themselves, in order to have no arms crossing.

Proof: When person a_k handshakes with person a_{k+d} their linked arms make a chord across the circular table, splitting the table into 2 halves. One halve contains the people $a_{k+1}, a_{k+2}, ..., a_{k+d-1}$, and the other half contains the remainding people. If 2 people from different halves were to handshake then their arms would cross that chord, which is a contradiction. □

Due to our lemma if person a_k handshakes with person a_{k+d}, it creates a group of $d-1$ people who have to handshake amongst themselves. For this to be possible $d-1$ must be even and d must be odd. Thus if person a_p handshakes with person a_q, the integers p and q must have opposite parity.

Let H_{2k} be the number of ways $k > 2$ pairs of people can handshake such that no arms cross, given that the $2k$ people are sat around a circular table. Consider person a_1, they must handshake with person a_{2m}, for integer $1 \leq m \leq k$. Then there are 2 groups of people that should handshake among themselves under the same constraints as before. One group is of size $2m-2$ and the other is of size $2k-2m$. Thus

$$H_{2k} = \sum_{m=1}^{k} H_{2m-2} \cdot H_{2k-2m}$$

It is easily verifiable that $H_0 = 1, H_2 = 1$ and from there one can show that $H_12 = 132$ by brute-force computation (or by using the fact that H_{2n} is the n^{th} catalan number $\frac{1}{n+1}\binom{2n}{n}$). ■